$$\mathbf{P}^2 = -\hbar^2 \frac{\partial^2}{\partial x^2},$$

and the Hamiltonian becomes

$$\mathbf{H} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}.$$

Finally, if we equate the left- and right-hand sides of the time-dependent Schrödinger equation, we get

$$i\hbar\frac{\partial\psi}{\partial t} = \frac{-\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2}. \tag{9.4}$$

This is the traditional Schrödinger equation for an ordinary nonrelativistic free particle. It is a particular kind of wave equation, but, in contrast to the previous example, waves of different wavelength (and momenta) move with different velocities. Because of this, the wave function *does not* maintain its shape. Unlike the zaxon wave function, it tends to spread out and fall apart. This is shown schematically in Fig. 9.2.

